Rural Hospital Risk Indicators: Predicting Closure Risk

Project Statement

Across the United States rural hospitals play a large role in providing healthcare access to smaller communities as they are often the main source of care. These hospitals are experiencing financial pressures due to lower patient volumes, high fixed costs, and a heavy reliance on reimbursement programs. As closures continue these trends raise growing concern about healthcare access and equity in rural communities. Research has shown that rural hospital closures have increased significantly over the past two decades, largely driven by financial sustainability and structural challenges (Cecil G. Sheps Center for Health Services Research, 2024).

The project's focus is to create an early warning system that flags hospitals that may be at risk of closure within the next one to two years. Our goal is not to perfectly predict closures but to identify patterns associated with financial and operational distress and translate those into practical ways to assess risk. The model is designed to function as a decision support tool to help policymakers and healthcare organizations prioritize monitoring and intervention efforts before closure occurs.

Methodology

We built a multi-year dataset using CMS Hospital Cost Reports and CMS Provider of Service files for the years 2013 to 2023, which included detailed hospital information on finances, utilization, and facility characteristics. We then linked these to the UNC Rural Hospital Closure dataset to identify historical rural hospital closures across the United States. One challenge during this process was that there was not a shared identifier across datasets. To resolve this, we created a matching approach based on hospital name, city, state, and address fields. This required standardizing text inputs and applying validation checks to improve match quality. This approach was not perfect but it enabled us to create a consistent longitudinal dataset that captures both financial performance and closure outcomes. The data was split using a time based approach where data prior to 2020 was used for training and data from 2020 onward was used for testing. This reflects how the model would be used in practice, where predictions are made on future data rather than on randomly mixed observations.

The final dataset includes approximately 26,000 hospital-year observations during the 2013-2023 time span. Closure events are a rare outcome with approximately 0.22% of observations happening within one year. Due to this we defined both one-year and two-year

closure targets to better capture short and medium term risk. To represent hospital performance in a meaningful way, we created a set of financial and operational ratios. These include profitability measures such as operating margin and total margin, efficiency measures such as cost-to revenue ratio, liquidity indicators such as current ratio and cash ratio, leverage measures such as debt ratio, and utilization indicators such as occupancy rate and average length of stay. We also incorporated payor mix and uncompensated care. Prior research has consistently shown that these types of financial and utilization measures are closely linked to hospital sustainability, particularly in rural settings (Holmes, George M et al., 2013).

We included both logistic regression and random forest models to balance interpretability and predictive performance. We began with logistic regression as it provides a transparent baseline and allows for straightforward interpretation of feature effects followed by random forest as it captures nonlinear relationships and interactions that are common in complex financial systems. Comparing these approaches also helps validate that observed patterns are consistent across different modeling techniques. Because closures are extremely rare, we also addressed class imbalance during modeling. Logistic regression was trained using class weighting to increase the importance of closure observations. The random forest model handles some imbalance through its structure but we further addressed this through threshold tuning rather than relying on default classification thresholds. We chose not to apply oversampling techniques, as maintaining realistic data distributions was important for this application.

Evaluation Strategy and Model Performance

Careful consideration is required to evaluate this problem as closure events are extremely rare. A model that predicts no closures would still achieve very high accuracy, making accuracy an unreliable metric. We instead focused on metrics that better capture performance in imbalanced settings, including ROC-AUC, precision-recall AUC (PR-AUC), recall, and precision. PR-AUC is especially important here because it provides a more realistic view of performance for rare events than ROC-AUC alone. We evaluated two prediction targets, closure within one year (primary model) and closure within two years (secondary model). The one year model serves as the primary focus because it aligns more closely with the goal of early intervention while the two year model provides additional context on medium term risk. For each target, we implemented both logistic regression as a baseline model and random forest model as the primary modeling approach.

The logistic regression model demonstrated moderate discriminatory ability, with a ROC-AUC of approximately 0.76 for the one-year prediction, but failed to identify meaningful numbers of closures while PR-AUC was effectively zero at standard thresholds. The two-year prediction performed slightly better in terms of PR-AUC but had lower ROC-AUC compared to the one-year

model suggesting there is better identification of rare closure events but weaker class separation. The random forest model improved performance for both timeframes. The one-year prediction reached an ROC-AUC of 0.87 and PR-AUC of about 0.05. Performance for the two-year prediction was similar, with higher PR-AUC but slightly lower ROC-AUC showing the same tradeoff pattern in the logistic regression model. While precision remains low due to the rarity of closures, both one and two-year prediction models demonstrate strong ability to rank hospitals by relative risk.

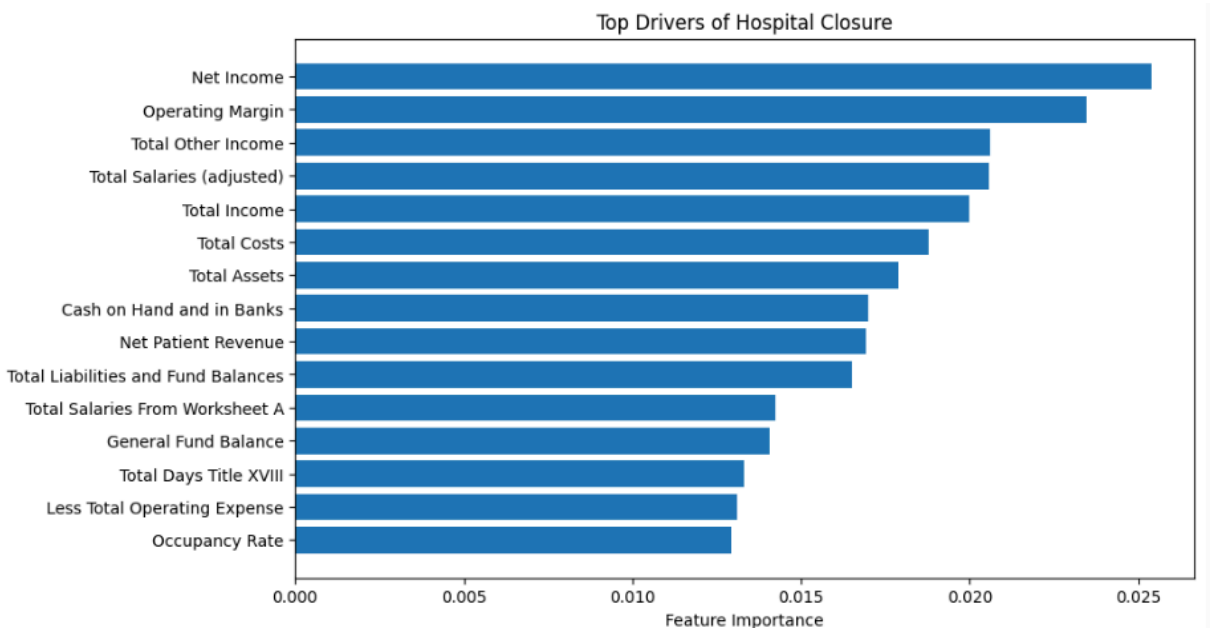
Both the one-year and two-year models showed predictive ability but the one-year model was selected as our main focus. A one-year prediction window aligns with the timeframe for policymakers and healthcare organizations to realistically intervene. By flagging a hospital as high-risk within the next year immediate monitoring, financial review, or targeted support can take place while a two-year prediction may be less precise in guiding decisions. For real world decision making, a one-year timeframe better fits where stakeholders need to act quickly to prevent closures. The one-year model is used as the primary tool for prioritizing at-risk hospitals and the two-year model provides additional context for longer term trends.

To make the model actionable, we implemented threshold tuning on the one-year model. By lowering the classification threshold recall increases but also increases the number of hospitals flagged. When the threshold is set to 0.15, the model captures approximately 60% of closures and flags about 10.5% of hospitals. This shows the tradeoff between identifying at-risk hospitals while maintaining a manageable number of flagged cases.

Model errors have real-world implications in this context. A false positive occurs when the model identifies a hospital as high-risk even though it does not ultimately close. This could result in additional monitoring or financial support being directed to a hospital that may not strictly need it. While this may create inefficiencies in resource allocation, it still provides some benefit by strengthening hospitals experiencing early signs of financial pressures. A false negative would be a more serious concern if the model fails to identify a hospital that is actually at risk of closure. For rural communities that already have limited access to care, this could mean that no intervention occurs resulting in reduced access for these individuals, longer travel times to receive care, and potentially worse health outcomes. Due to resources being limited to support hospitals, these errors also create an important equity tradeoff. If resources are provided to hospitals that are incorrectly identified as high-risk, hospitals that are actually vulnerable may not receive the support they need. This is why the model is intentionally tuned to prioritize recall, this ensures that a larger number of at-risk hospitals are identified.

Key Drivers of Hospital Closure

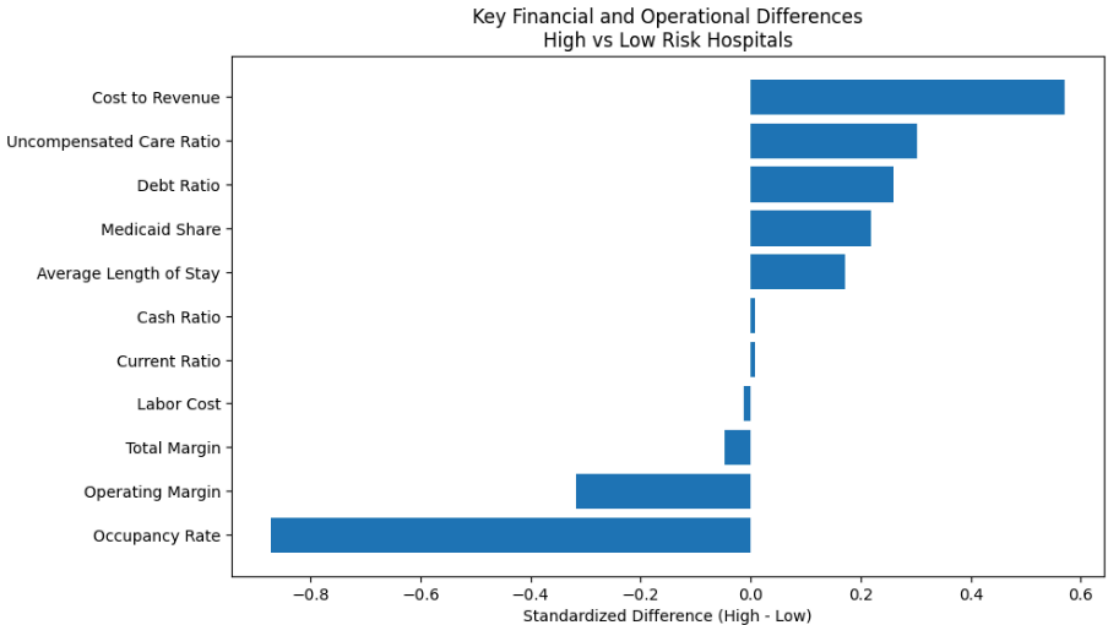
The model's feature importance results provide insight into the main drivers of closure risk. The *Top Drivers of Hospital Closures* graph shows that financial variables dominate the model, with net income, operating margin, total income, and total costs are among the most important features. This aligns closely with prior research, which has found that weak financial performance is one of the strongest predictors of rural hospital closure (Holmes, George M et al., 2013). What stands out is that both revenue and cost related variables appear together as top drivers. This suggests that closure risk is not just the result of low revenue but an imbalance between revenue and cost structure, therefore hospitals that are not able to adjust costs as revenue declines may be more vulnerable. This points to a larger structural challenge in rural healthcare as many hospitals face high fixed costs but limited ability to increase patient volume.



These findings are consistent across modeling approaches. Financial indicators such as margins, cost structure, and revenue levels appear as key drivers in both the random forest feature importance results and the logistic regression coefficient analysis. The survival analysis further reinforces these results as hospitals with weaker financial positions show a faster rate of closure over time. This consistency increases our confidence that the model is identifying meaningful and stable signals rather than patterns specific to a single modeling method. This suggests that early warning efforts should focus heavily on financial indicators, in particular those that capture cost pressures relative to revenue and monitoring these signals may provide earlier visibility into risk before closures occur.

Financial and Operational Differences

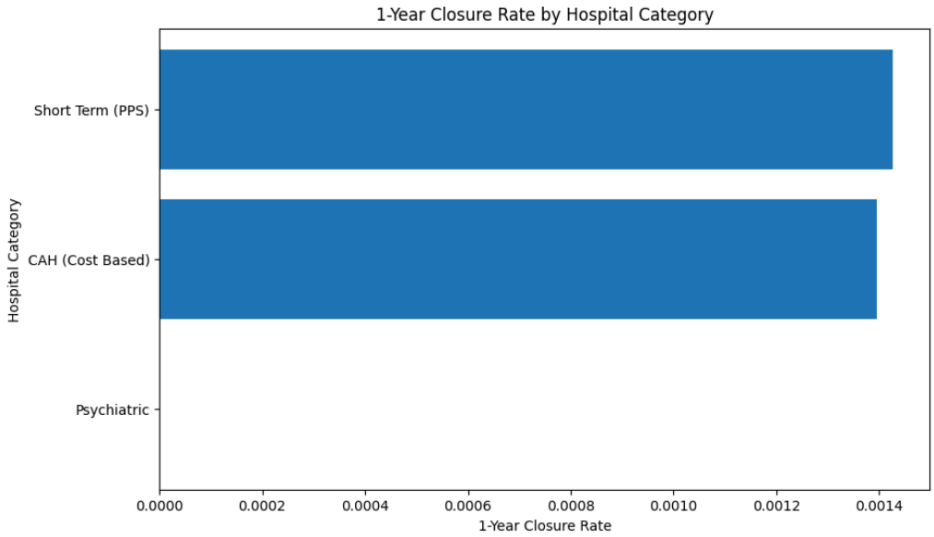
The *Key Financial and Operational Differences* graph shows that high-risk hospitals tend to have higher cost to revenue ratios, high uncompensated care, higher debt levels, and greater reliance on Medicaid. These findings are consistent with prior studies that highlight financial distress and payor mix as key drivers of closure risk (Kaufman et al., 2016). These patterns taken together suggest that high-risk hospitals are operating with structurally weaker revenue streams while also carrying greater financial obligations. A higher share of Medicaid patients and uncompensated care ratio reduces the amount of revenue hospitals receive even when there is demand for care. This creates a situation where hospitals are providing necessary services but are not being fully compensated for them.

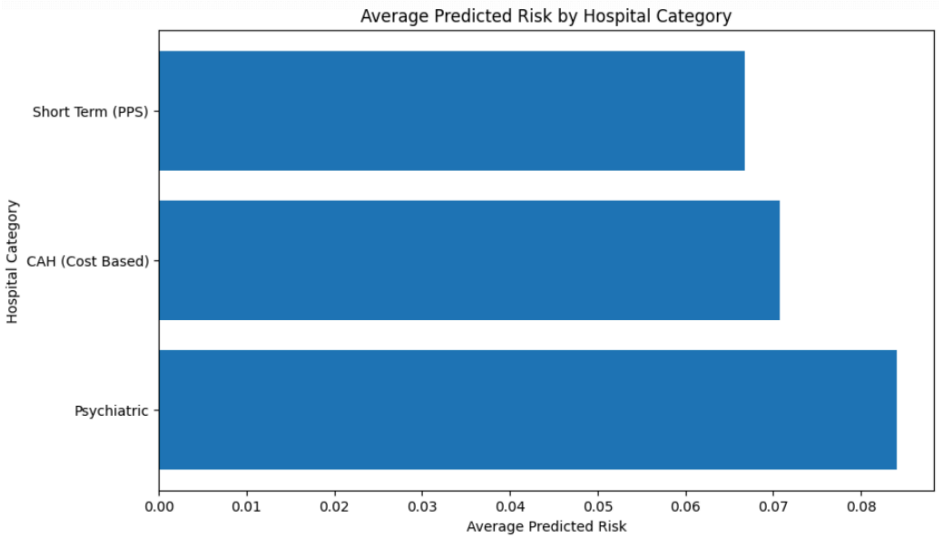


Operational differences reinforce this dynamic. High-risk hospitals show as having lower occupancy rates and weaker liquidity, indicating both demand challenges and limited financial flexibility. Lower occupancy suggests that hospitals are not able to fully utilize their capacity and lower liquidity reduces their ability to absorb short term financial shocks. This combination of financial pressure and operational constraints creates a compounding effect. Hospitals that are both underutilized and financially constrained are less able to adapt when conditions worsen which makes them more vulnerable to closure. Interventions may need to address both financial structure and utilization patterns and not just on a single issue.

Reimbursement Type and Closure Risk

The reimbursement structure of a hospital can provide additional information for understanding how policy and funding models influence risk. Short-term acute care hospitals are reimbursed based on a set amount dependent on the type of patient treated, regardless of the actual cost of care. Critical Access Hospitals (CAHs) are reimbursed based on their actual costs, which is intended to support smaller rural hospitals. The *1-Year Closure Rate by Hospital Category* graph shows that short-term and CAHs have similarly observed closure rates. However the *Average Predicted Risk by Hospital Category* graph indicates that CAHs have slightly higher predicted risk compared to short term.



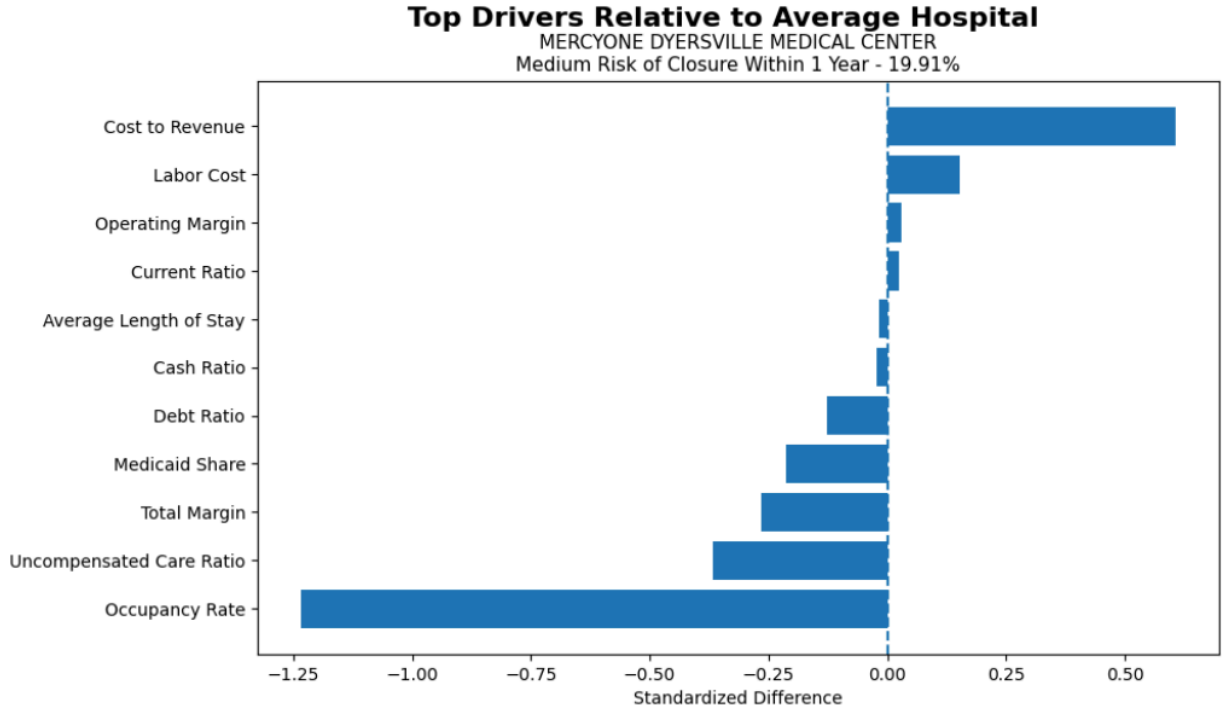


CAHs operate under a cost-based reimbursement system designed to provide additional financial support to rural providers but the model suggests that these hospitals may still

experience greater underlying financial vulnerability. This indicates that reimbursement structure by itself is not sufficient to ensure stability. Other factors such as low patient volume, location, or high fixed costs also play a large role in closure risk, therefore from a policy perspective addressing rural hospital closures should include a wider range of strategies. Combining reimbursement adjustments with other strategies that address utilization, cost structure, or regional healthcare demand is a stronger approach.

Hospital Level Interpretation: Drivers Relative to Average

The *Top Drivers Relative to Average Hospital* graph shows how individual hospital characteristics compare to the overall average and how these differences contribute to the risk prediction provided in the sub title. In the below example, a high cost to revenue ratio and elevated labor costs are the primary drivers of risk both significantly above average. At the same time the occupancy rate is below average, meaning the hospital's costs and labor costs are high while the actual patients being seen is low. Additional factors such as high debt levels and low operating margins further contribute to the hospital's risk.

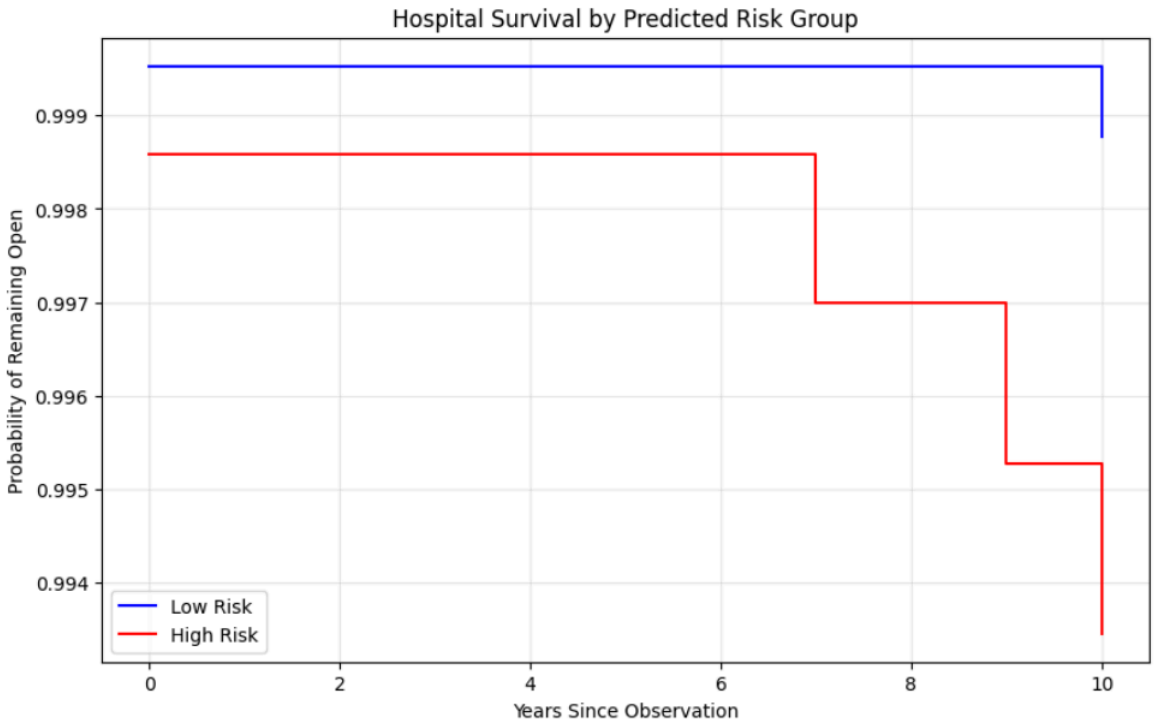


What makes this analysis valuable is its ability to move beyond just predicting hospital risk but provide explanation on the specific factors driving the risk. This makes the results more actionable and easier to interpret for decision making and this type of output can support more targeted interventions. For example, a hospital with high labor costs may require operational

adjustments while a hospital with low occupancy would focus on strategies aimed at increasing patient volume. The model can become a more tailored approach rather than applying the same solution across the board and interventions can be prioritized and targeted effectively.

Survival Analysis and Validation

To support the classification models, we created a survival analysis using the Kaplan-Meier estimator to examine how closure risk evolves over time. Hospitals were grouped into high and low-risk categories based on predicted probabilities from the random forest model tracked to observe closure timing. The Hospital Survival by Predicted Risk Group graph shows a clear separation between these groups. High-risk hospitals experience a larger decline in survival probabilities over time and close at a faster rate. Low-risk hospitals however remain relatively stable and have very high probabilities of staying open through the observed period.



This separation provides strong validation that the model is capturing meaningful differences in risk. If the model was not informative, we would expect the survival curves to overlap more closely. We instead observe a gap that suggests that the model's risk classifications reflect real differences in long-term outcomes, consistent with prior research (Holmes, George M et al., 2013). The survival analysis reinforces that hospitals identified as high-risk are not only more likely to close, but tend to do so sooner. This supports the model's use as an early warning tool for identifying and prioritizing at-risk hospitals.

Translation of Results and Practical Use

The model demonstrates strong performance in distinguishing high-risk hospitals but it is important to interpret its results in context. Due to the rarity of closures, precision remains low which means that many hospitals flagged as high-risk will not actually close. This does not indicate that the model is ineffective, it instead suggests the model is best understood as a prioritization tool rather than a definitive predictor. Its primary value lies in its ability to rank hospitals by relative risk and identify a subset that needs closer attention which is supported by the survival analysis.

This model helps policymakers and healthcare organizations focus on hospitals that are more at-risk. For example, the top 10-15% of hospitals flagged by the model could be tracked more closely and reviewed in more detail. It can also be used to look at how changes in reimbursement, cost structure, or utilization affect risk over time. The model should be used as part of a broader decision making framework and is meant to support existing expertise.

Broader Impacts

Identifying at-risk hospitals earlier could help maintain access to care and improve outcomes for communities with few or no other alternatives for emergency or inpatient services. Risk scores are useful for prioritizing attention but are not meant as a perfect prediction of closures. As mentioned earlier, the model should be used to support more proactive and equitable resource allocation and should not be applied strictly as it could unintentionally direct support away from communities whose needs are less visible in the data.

Conclusion

This project demonstrates that it is possible to to build a meaningful early warning system for rural hospital closures utilizing publicly available data. It successfully identifies patterns associated with increased risk and provides a practical framework for prioritizing interventions. The biggest contribution of this work is its ability to translate complex financial and operational data into actionable insights and supports more informed decision making in a critical area of healthcare access.

Statement of Work

Team members shared the responsibility for project design, interpretation, and written report.

- Kostas Boutsikakis led data acquisition and preprocessing.
- McKenzie Boutsikakis led modeling and evaluation.

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